AP Computer Science Principles - Unit 6 Data Analysis Quiz

1. (DOK 1) Which step of the data science process focuses on obtaining data?
   A. Plan
   B. Collect
   C. Visualize
   D. Recommend

2. (DOK 1) What is the purpose of data visualization?
   A. Store data
   B. Model data to see patterns
   C. Delete duplicate data
   D. Collect survey responses

3. (DOK 1) Which stage removes incorrect, incomplete, or duplicate data?
   A. Collect
   B. Recommend
   C. Clean
   D. Plan

4. (DOK 1) According to the lesson, data analysis includes collecting, cleansing, and _____ data.
   A. encrypting
   B. sharing
   C. visualizing
   D. compressing

5. (DOK 2) A student creates a survey that only iPhone users can access. Which concern should be addressed?
   A. Visualization error
   B. Data collection bias
   C. File formatting
   D. Recommendation stage

6. (DOK 2) Why did the lesson use the GED VIZ tool?
   A. To write computer code

B. To visualize data and identify patterns
C. To store passwords
D. To create video games

7. (DOK 2) Why is the planning stage important before collecting data?
A. It identifies goals, scope, and collection methods
B. It automatically cleans data
C. It creates charts
D. It eliminates bias completely

8. (DOK 3) A graph shows students who sleep more tend to earn higher grades. What is the best conclusion?
A. Sleep definitely causes higher grades
B. The graph proves causation
C. There may be a correlation, but more research is needed
D. The data is invalid

9. (DOK 3) A research team has duplicate, incomplete, and inconsistent survey responses. What should they do before creating visualizations?
A. Make recommendations
B. Gather unrelated data
C. Clean and organize the data
D. Publish results immediately

10. (DOK 4) A school wants to study the impact of social media on student sleep. Which plan best follows the entire data science process?
A. Survey students, clean responses, create visualizations, analyze findings, and make recommendations
B. Create a chart first and then ask questions
C. Collect random data and publish it
D. Make recommendations without collecting data